

L8: Knowledge Check

1 Multiple choice 1 point

Which of the following statements regarding the three approaches to identify communities in a dynamic network is/are **TRUE**:

- ☐ The evolution of network community structure is done by computing communities separately at each time step.
- ☐ The dual-objective approach leads to the formation of communities with lower modularity at time $T+1$ than the approach where the communities are separately computed at each time step.
- ☐ The global network is formed by classifying the nodes and links at each time step as local and the links between networks at various time steps as global.
- ☐ The dual-objective approach leads to the formation of the smoothest results as it considers the communities formed at all the previous steps indirectly.

2 Multiple answer 1 point

Which of the following statements about the CFinder algorithm and random networks is/are **FALSE**:

- ☐ The critical density threshold increases with the value of k (k for k -cliques).
- ☐ The critical density threshold is the highest density of a random network that makes the formation of k -cliques likely.
- ☐ There are never any k -cliques observed below the critical density threshold.
- ☐ If a network is denser than the critical threshold, CFinder will probably detect many communities even if the network is random.

3 Multiple answer 1 point

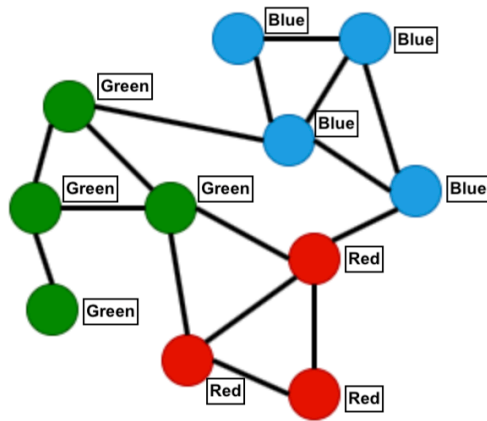
Which of the following statements about the LFR benchmark is/are **TRUE**:

- ☐ The LFR benchmark generates a network in which all communities have the same size.
- ☐ The only difference between LFR and GN networks is how they model the degree distribution.
- ☐ LFR generates networks in which both the node degrees and community sizes follow power-law distributions.
- ☐ LFR networks can also have overlapping communities.

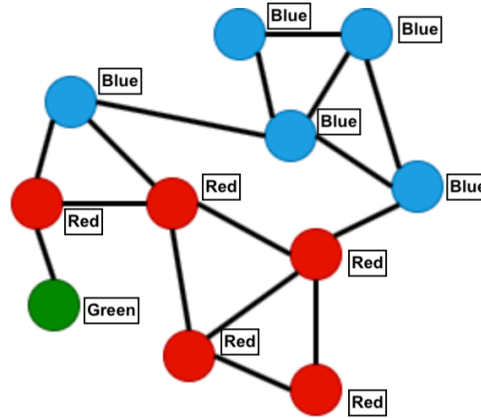
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Multiple choice 1 point

Compute the confusion matrix for the two given partitions and find the missing entries in cells (1,2),(2,1),(2,3),(3,2) in the same order.



Partition : A



Partition : B



Row,Col : A,B	Red	Green	Blue
Red	3		0
Green		1	
Blue	0		4

- ☐ 0,1,2,0
☐ 3,1,2,3
☐ 3,2,1,3
☐ 0, 2,1,0

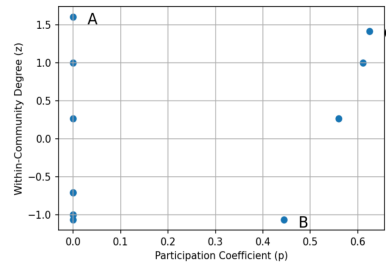
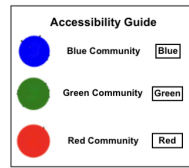
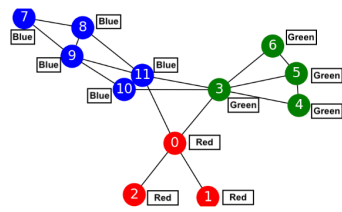
5

Multiple answer 1 point

Which of the following statements about Delta-MAPS, as a method to identify spatially-contiguous clusters of grid-cells based on correlations, is/are TRUE:

- ☐ The objective of Delta-MAPS is to identify the maximum number of minimal-size “domains” in the given data.
☐ A Delta-MAPS domain can be thought of as a spatially contiguous community in which the nodes represent grid points and the edges represent temporal correlations between the corresponding grid points.
☐ The Delta-MAPS algorithm computes one domain in each iteration. In a given iteration, the objective is to compute the largest possible domain subject to the constraint imposed by the parameter δ .
☐ Delta-MAPS domains can be overlapping.

The left plot shows a network, while the right plot shows one point for each node where the x-coordinate is the Participation Coefficient and the y-coordinate is the Within-Community Degree. We know that the points A, B, C at the right plot correspond to the nodes 0, 9, 10 at the network. Which of the following is the most likely correspondence between points A, B, C and nodes 0, 9, 10?



- ☐ A - 10, B - 9, C - 0
- ☐ A - 9, B - 0, C - 10
- ☐ A - 9, B - 10, C - 0
- ☐ A - 0, B - 10, C - 9